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a fabricated fact that contradicts the provided context which BenHalluEval is designed to detect.

Hallucination evaluation has been explored in different languages such as English (Li et al., 2023) Arabic (Alansari and Luqman, 2025), Hindi and Turkish (Abdaljalil et al., 2026). Bengali, being the sixth most spoken language in the world (Sani et al., 2026), remains under-explored. Dedicated pretrained models (Bhattacharjee et al., 2022), multilingual benchmarks (Hasan et al., 2021), and task-specific datasets (Clark et al., 2020; Khan et al., 2023; Paul et al., 2025) already exist for Bengali, yet no benchmark measures hallucination in LLMs on both native Bengali and Bangla-English code-mixed input.

To fill this gap, we introduce **BenHalluEval**, a hallucination evaluation framework for Bengali covering four tasks: Generative Question Answering (GQA), Bangla-English Code-Mixed QA, Summarization, and Reasoning. We use GPT-5.4¹ to generate hallucinated candidates, evaluate seven LLMs across three model categories, and systematically evaluate chain-of-thought prompting as a mitigation strategy across all model-task combinations. Our main contributions are:

- We present **BenHalluEval**, a multi-task hallucination evaluation framework for LLMs on Bengali and Bangla-English inputs, covering Generative Question Answering (GQA), Bangla-English code-mixed QA, summarization, and mathematical reasoning.
- We construct 12,000 hallucinated candidates using GPT-5.4 across task-specific hallucination categories, including a Bangla-English code-mixed setting, and evaluate them alongside ground-truth instances using an LLM-as-judge framework.
- We introduce **BenHalluScore** to jointly measure false positives and missed hallucinations, benchmark seven LLMs across three model categories, and systematically evaluate chain-of-thought prompting across all seven models and three tasks, showing the task-dependent patterns of improvement and degradation..

¹<https://openai.com/index/introducing-gpt-5-4/>

2 Related Work

Hallucination Evaluation Benchmarks.

TruthfulQA, one of the earliest works on hallucination evaluation (Lin et al., 2022), showed that model scale alone does not guarantee factual accuracy. FActScore advanced this by breaking long-form outputs into atomic claims for fine-grained precision measurement (Min et al., 2023). HalluEval established the sampling-then-evaluation paradigm across QA, dialogue, and summarization (Li et al., 2023). FaithBench (Ping and Durrett, 2024) and HalluLens (Bang et al., 2025) subsequently refined faithfulness annotations and guarded against benchmark leakage. MedHallu extended this to the clinical domain, showing how domain specificity amplifies the impact of hallucination errors (Pandit et al., 2025). A shared limitation across these benchmarks is labelling bias where LLM judges tend to default to one verdict class regardless of content, a problem none of them directly measure (Li et al., 2023; Adib et al., 2025).

Low-Resource Hallucination and Bengali NLP.

Hallucination rates scale inversely with pretraining data availability, with the gap widening in open-ended generation (Chataigner et al., 2024; Islam et al., 2025). AraHalluEval (Alansari and Luqman, 2025) evaluated Arabic-capable LLMs using fine-grained hallucination indicators, and Halluverse-M³ (Abdaljalil et al., 2026) extended multilingual evaluation to English, Arabic, Hindi, and Turkish. Neither benchmark incorporates dual-track calibration or examines code-mixed input. More recent work covers multi-domain understanding via BnMMLU (Joy and Shatabda, 2025), reasoning via MathMist (Sobhani et al., 2025), and summarization consistency via BanglaSummEval (Rafid et al., 2026). Bangla-English code-mixed usage is well-documented (Alam et al., 2025; Sheth et al., 2025; Alam et al., 2026), yet hallucination behaviour on Bangla-English input remains unexamined.

Mitigation Strategies. Chain-of-thought prompting (Wei et al., 2022) improves multi-step inference but can paradoxically obscure hallucination cues, making its benefit task- and model-dependent (Cheng et al., 2025). (Cheng et al., 2025) show that CoT reasoning

can shift a model’s prediction patterns without improving its ability to distinguish hallucinated from correct content, with gains limited to tasks that require structured, verifiable intermediate steps.

No prior work has built a hallucination evaluation framework for Bangla. BenHalluEval addresses this by covering four tasks: GQA, Bangla-English Code-Mixed, QA, Summarization, and Reasoning under a unified dual-track evaluation protocol.

3 Methodology

BenHalluEval is constructed through a three-stage pipeline: (i) seed dataset selection and sample filtering, (ii) hallucinated sample generation using GPT-5.4, and (iii) hallucination detection evaluation using seven LLMs. Figure 2 provides a full system overview. Each task has a task-specific selection strategy and hallucination categorization; the evaluation protocol is unified across all four tasks.

3.1 Stage 1: Seed Dataset Selection and Sample Filtering

Generative Question Answering. We use TyDiQA-GoldP (Bengali) (Clark et al., 2020), a dataset of 2,509 question–passage–answer triples. To select the samples that are more prone to hallucination, we prompted three models zero-shot with only the question, without the passage: Qwen2.5-14B², Gemma-2-9B³, and DeepSeek-R1-14B⁴. A Qwen2.5-32B-Instruct⁵ judge scores each response as correct or incorrect. Samples that all three models answer incorrectly form the hard subset, yielding 1,313 instances (52.33%), from which 1,000 are randomly selected as the GQA seed dataset. To verify judge reliability, 100 instances were manually annotated and compared against the judge verdicts, finding high agreement. Table 1 shows per-model zero-shot accuracy.

Bangla-English Code-Mixed QA. Bangla-English code-mixed text is widely used by Bengali speakers on social media

Model	Incorrect	Err. (%)
DeepSeek-R1-14B	1,747	30.37
Gemma-2-9B	1,668	33.54
Qwen2.5-14B	1,948	22.36
All 3 incorrect	1,313	52.33

Table 1: Zero-shot accuracy of three models on TyDiQA-GoldP (Bengali, 2,509 instances).

and messaging platforms (Alam et al., 2025; Sheth et al., 2025; Alam et al., 2026). To evaluate hallucination detection in this setting, we take the same 1,000 GQA samples and convert the questions from Bengali to Bangla-English code-mixed using GPT-5.4 (see Appendix C.5). Original answers are kept in Bengali. No extra filtering is needed, since these samples are already the hard subset from the GQA selection. To verify conversion quality, we manually reviewed 100 randomly selected samples. All authors are native Bengali speakers, and the reviewed samples were found to be natural and linguistically appropriate.

Summarization. We use BanglaCHQ-Summ (Khan et al., 2023), a dataset of 1,800 Bengali consumer health query–summary pairs. We filter the samples based on semantic similarity using BERTScore F1. The same three models generate zero-shot summaries for all 1,800 samples, which are then scored against the original summaries using BERTScore F1 (Zhang et al., 2020). The average BERTScore F1 is computed per sample across all three models. The 1,000 samples with the lowest average scores are selected, as these are the samples that all three models struggle most to summarize faithfully.

Reasoning. We use SOMADHAN (Paul et al., 2025), a dataset of 8,792 Bengali math word problems with manually written chain-of-thought (CoT) solutions. Samples are ranked by the length of their reasoning chain. The 500 longest and 500 shortest reasoning chains are selected as the two sampling pools. For each of the five error types, 100 samples with the longest reasoning chains and 100 with the shortest are selected, yielding 1,000 samples in total (200 per type). To ensure equal coverage, samples are picked alternately from the

²<https://huggingface.co/Qwen/Qwen2.5-14B>

³<https://huggingface.co/google/gemma-2-9b>

⁴<https://ollama.com/library/deepseek-r1:14b>

⁵<https://huggingface.co/Qwen/Qwen2.5-32B-Instruct>

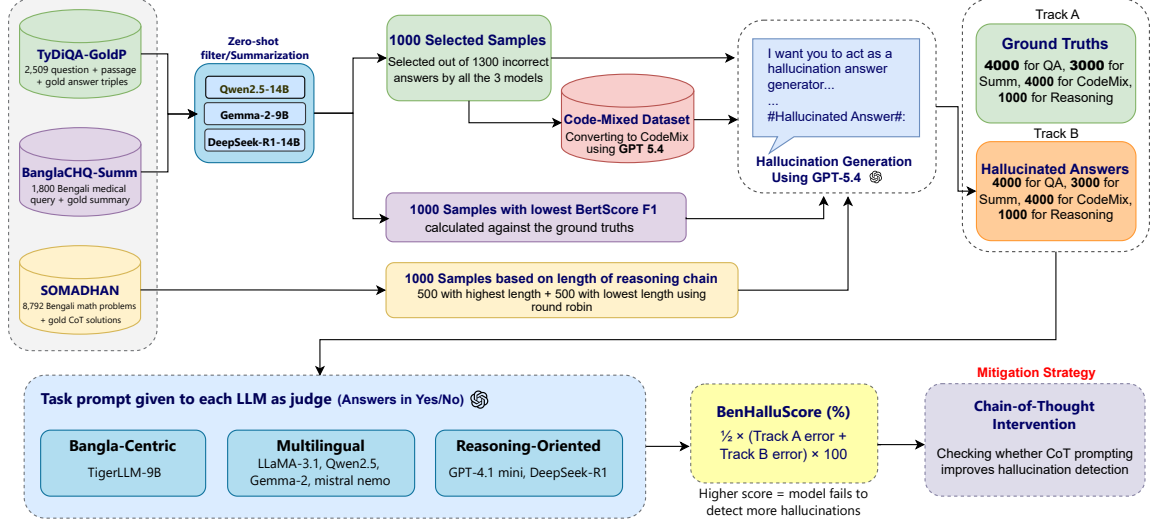


Figure 2: BenHalluEval system pipeline. Four tasks (GQA, Code-Mixed QA, Summarization, Reasoning) share a common evaluation stage. Track A evaluates ground-truth correct answers; Track B evaluates hallucinated candidates.

long and short pools in a fixed round-robin order, so that each error type receives an equal number of long and short reasoning chains.

3.2 Stage 2: Hallucinated Sample Generation

All hallucinated samples are generated using GPT-5.4. Each prompt specifies the hallucination type, includes an example of hallucination, and formats the target sample with field markers.

GQA Hallucination Generation. Four hallucination types are applied to each of the 1,000 GQA seed samples, one API call per type (full prompt is in Appendix C.2). This generates 4,000 hallucinated candidates in total:

- **Factualness:** Fabricates a concrete fact such as a name, date, or place that contradicts the context.
- **Comprehension:** Misunderstands the question’s context or intention and answers a different question.
- **Specificity:** Answers at an inappropriate level of detail, either too vague or too specific.
- **Inference:** Draws a conclusion that cannot be logically inferred from the given information.

Bangla-English Code-Mixed QA Hallucination Generation. The same four hallucination types used for GQA are applied to the 1,000 Bangla-English Code-mixed seed samples. The generation process mirrors the GQA pipeline exactly, with one difference: both the prompt and the generated hallucinated answers are in Bangla-English (Roman-script Bengali) rather than native Bengali script (full prompt is in Appendix C.2).

Summarization Hallucination Generation. Three hallucination types are applied to each of the 1,000 summarization seed instances, generating 3,000 hallucinated summaries in total:

- **Fabricated Content:** Introduces a detail that does not exist anywhere in the source document.
- **Non-Factual Addition:** Inserts a plausible but unverifiable claim not supported by the source.
- **Direct Contradiction:** Reverses information that is explicitly stated in the source.

Full prompt is in Appendix C.3.

Reasoning Hallucination Generation. We apply five error types, one per sample, selecting 1,000 hallucinated CoT chains. GPT-5.4 is prompted to modify exactly one reasoning step per instance and return output as a

I want you act as a hallucination answer generator. The answer should be given in BANGLA . Given a question, right answer, and related knowledge, your objective is to write a hallucinated answer that sounds plausible but is factually incorrect. You SHOULD write the hallucinated answer using the following method:
1. You are trying to answer a question but there is a factual contradiction between the answer and the knowledge. You can fabricate some information that does not exist in the provided knowledge. (Demonstrations)
2. You are trying to answer a question but you misunderstand the question context and intention. (Demonstrations)
3. You are trying to answer a question but the answer is too general or too specific to answer the question at an appropriate level of specificity. (Demonstrations)
4. You are trying to answer a question but the answer cannot be inferred from the knowledge. You can incorrectly reason with the knowledge to arrive at a hallucinated answer. (Demonstrations)
You should try your best to make the answer become hallucinated. #Hallucinated Answer# can only have about 5 more words than #Right Answer# . #Knowledge#: <i><insert the related knowledge/context></i> #Question#: <i><insert the question></i> #Right Answer#: <i><insert the right answer to the question></i> #Hallucinated Answer#: Generate

Table 2: Hallucination generation prompt for the GQA task. Four methods correspond to factual contradiction, comprehension mismatch, inappropriate specificity, and incorrect inference errors. Each instance generates one hallucinated answer per method under controlled constraints. Full prompt with Bengali demonstrations is in Appendix 9.

structured JSON object (full prompt in Appendix C.4):

- **Arithmetic Slip:** Silently alters a single numerical computation by a small amount.
- **Formula Misapplication:** Substitutes a related but incorrect mathematical operation.
- **Invalid Deduction:** Injects a logically unsupported transition between two reasoning steps.
- **Hallucinated Intermediate Fact:** Introduces a fabricated assumption mid-chain as if it were given.
- **Semantic Drift:** Reinterprets the question mid-reasoning so the final answer addresses a different goal.

Task	Types	Seed	Hallu.	Gold
GQA	4	1,000	4,000	1,000
Code-mix	4	1,000	4,000	1,000
Summarization	3	1,000	3,000	1,000
Reasoning	5	1,000	1,000	1,000
Total			12,000	4,000

Table 3: BenHalluEval benchmark statistics. “Hallu.” = hallucinated candidates generated by GPT-5.4; “Gold” = ground-truth correct instances (Track A evaluation).

3.3 Stage 3: Hallucination Detection Evaluation

Evaluation tracks. Evaluating only on hallucinated candidates risks rewarding models that blindly predict “Yes” for every input, while evaluating only on ground-truth samples risks rewarding models that always predict “No”. To get a more accurate picture of model behaviour, we evaluate each task on two separate tracks. **Track A** presents 1,000 ground-truth correct answers, where the expected verdict is “No” (not hallucinated). **Track B** presents the hallucinated candidates, where the expected verdict is “Yes”. This separation allows hallucination detection rate and false-positive rate to be measured independently.

To measure performance across both tracks jointly, we introduce **BenHalluScore** (BHS). A good model must both detect hallucinated content and avoid flagging correct answers as hallucinated. Neither track alone captures this. BenHalluScore penalises both failure modes equally, so a higher score indicates worse overall calibration. Formally:

$$\text{BenHalluScore} = \frac{1}{2} \left(\frac{W_A}{N_A} + \frac{W_B}{N_B} \right) \times 100\% \quad (1)$$

where W_A and W_B are the number of wrong verdicts on Track A and Track B respectively,

N_A and N_B are the total instances per track, and the factor $\frac{1}{2}$ normalises the combined error to the $[0, 100]$ range. A score of 0% indicates perfect calibration on both tracks; 50% indicates random or uniformly biased behaviour; 100% indicates perfectly wrong predictions across both tracks.

4 Experimental Setup

Models. We evaluate seven LLMs spanning three categories (Table 5). The reasoning-oriented category comprises DeepSeek-R1-14B and GPT-4.1 mini⁶. The multilingual category includes Qwen2.5-32B-Instruct, Gemma-2-27B⁷, Mistral-nemo-12B⁸, and LLaMA-3.1-8B-Instruct⁹. The Bangla-centric category is represented by TigerLLM-9B (Raihan and Zampieri, 2025).

Inference Settings. All models produce binary verdicts (*yes/no*) at temperature = 0, with a maximum output length of 1,024 tokens. Each instance is presented with its associated context - the passage for GQA and Bangla-English code-mixed QA, the source article for summarisation, and the problem statement for reasoning. GQA, summarisation, and reasoning prompts are issued in Bengali and the code-mixed prompts are in Bangla-English. GPT-4.1 mini is accessed via the OpenAI API. All remaining models are served locally using Ollama on a PC equipped with 128 GB RAM and an NVIDIA RTX 3090 GPU (24 GB VRAM). Full evaluation prompts are provided in Appendix D.

5 Results Analysis

5.1 Generative Question Answering

Table 4 shows GQA results under the dual-track BenHalluScore. GPT-4.1 mini achieves the best BenHalluScore (15.56%), maintaining low errors on both tracks (A-err. 13.90%, B-err. 17.23%) - the only model that performs well simultaneously on both. DeepSeek-R1-14B scores 38.60%, with a notably high

Track A error of 48.00%, meaning it incorrectly flags nearly half of all correct answers as hallucinated. Mistral-nemo-12B scores the worst (53.84%), driven almost entirely by its 87.70% Track A error - it flags nearly every ground-truth answer as hallucinated. LLaMA-3.1-8B shows the opposite pattern, predicting hallucination on 81.63% of Track B instances while maintaining a low Track A error (13.80%), reflecting opposing systematic biases rather than content-sensitive judgements. Notably, TigerLLM-9B, the only Bengali-centric model evaluated, achieves the second-best BenHalluScore on GQA (28.28%), outperforming significantly larger multilingual models including Qwen2.5-32B, Gemma-2-27B, and even the reasoning-oriented DeepSeek-R1-14B. This suggests that targeted Bengali pretraining provides a meaningful calibration advantage on native-script input despite the model’s smaller parameter count.

5.2 Bangla-English Code-Mixed QA

Bangla-English code-mixed results differ from GQA for several models (Table 4), showing that script condition interacts with model behaviour. GPT-4.1 mini remains the best (BHS 21.90%), and Mistral-nemo-12B (55.42%) and LLaMA-3.1-8B (45.93%) retain their characteristic biases in the Roman-script condition. However, Qwen2.5-32B, Gemma-2-27B, and DeepSeek-R1-14B improve sharply on Bangla-English code-mixed relative to GQA (by 22.85, 18.47, and 11.66 BHS points respectively), driven primarily by lower Track A errors. A plausible explanation is that Roman-script input shifts these models toward their stronger English-anchored representations, producing more consistent verdicts than native Bengali script - the opposite of the intuitive expectation that code-mixing would degrade performance.

5.3 Summarization

Across all models, BenHalluScores on summarization are notably lower than on GQA (Table 4), suggesting that access to a full source article provides stronger contextual basis for factual consistency judgements. Qwen2.5-32B achieves the lowest (best performing) BenHalluScore (7.72%), with low errors on

⁶<http://developers.openai.com/api/docs/models/gpt-4.1-mini>

⁷<https://huggingface.co/google/gemma-2-27b>

⁸<https://mistral.ai/news/mistral-nemo>

⁹<https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>

Model	GQA			Code-Mixed QA			Summarization			Reasoning		
	A-err.	B-err.	BHS	A-err.	B-err.	BHS	A-err.	B-err.	BHS	A-err.	B-err.	BHS
DeepSeek-R1-14B	48.00	29.20	38.60	40.10	13.78	26.94	17.60	6.73	12.17	19.90	16.30	18.10
GPT-4.1 mini	13.90	17.23	15.56	24.00	19.80	21.90	8.20	11.37	9.79	31.70	21.10	26.40
Qwen2.5-32B	27.90	65.80	46.85	17.40	30.60	24.00	4.70	10.73	7.72	24.80	28.80	26.80
Gemma-2-27B	16.40	78.50	47.45	19.80	38.15	28.98	4.20	31.40	17.80	5.50	72.60	39.05
Mistral-nemo-12B	87.70	19.98	53.84	85.90	24.93	55.42	70.50	9.87	40.19	1.60	97.00	49.30
LLaMA-3.1-8B	13.80	81.63	47.71	13.80	78.05	45.93	31.80	25.77	28.79	17.40	89.20	53.30
TigerLLM-9B	18.40	38.17	28.28	25.80	37.28	31.54	4.80	47.53	26.17	21.40	85.50	53.45

Table 4: BenHalluScore results across all four tasks and seven models (all values in %). A-err.: Track A error rate on ground-truth instances. B-err.: Track B error rate on hallucinated candidates. green cell = best (lowest) BHS per task; red cell = worst (highest) BHS per task. Full per-task results with raw counts in Appendix E.

Category	Model	Params
Reasoning	DeepSeek-R1-14B	14B
	GPT-4.1 mini	—
Multilingual	Qwen2.5-32B-Instruct	32B
	Gemma-2-27B	27B
	Mistral-nemo-12B	12B
	LLaMA-3.1-8B-Instruct	8B
Bangla-centric	TigerLLM-9B	9B

Table 5: Seven LLMs evaluated in BenHalluEval, grouped by model category.

both tracks (A-err. 4.70%, B-err. 10.73%), while DeepSeek-R1-14B ranks second (12.17%) despite recording the lowest Track B error (6.73%). Mistral-nemo-12B again scores worst (40.19%), driven by a 70.50% Track A error. TigerLLM-9B shows the opposite pattern - a 47.53% Track B error against a 4.80% Track A error (BHS 26.17%) - indicating that labelling bias persists even when the model has access to sufficient context.

5.4 Reasoning

Among all four tasks, reasoning shows the widest spread of BenHalluScore, ranging from 18.10% to 53.45% (Table 4). DeepSeek-R1-14B achieves the best and most balanced result (A-err. 19.90%, B-err. 16.30%, BHS 18.10%). At the other end, Mistral-Nemo-12B predicts “No” for nearly every instance regardless of content (A-err. 1.60%, B-err. 97.00%, BHS 49.30%), a pattern indicating uniform label collapse. LLaMA-3.1-8B and TigerLLM-9B occupy the opposite extreme with Track B er-

rors of 89.20% and 85.50% respectively, resulting in the worst BenHalluScores (53.30% and 53.45%). The concentration of models near or above 50% BenHalluScore indicates that reasoning hallucination in Bengali is particularly difficult to detect.

5.5 Chain-of-Thought Mitigation

Chain-of-thought (CoT) prompting has been shown to improve reasoning and reduce hallucination in LLMs by extracting step-by-step explanations before a final answer (Wei et al., 2022). We apply CoT to all seven evaluated models across GQA, Summarization, and Reasoning, resulting in 21 model-task combinations (Table 6). This provides a systematic view of CoT’s effect across model categories, scales, and task types. Figure 4 visualises BenHalluScore before and after CoT across all seven models.

Across the 21 model-task combinations, CoT results in a lower BenHalluScore in 9 instances while worsening performance in 12, confirming that step-by-step prompting does not consistently improve hallucination discrimination in Bengali. Genuine, balanced improvements, where both Track A and Track B errors either hold or decrease, are rare, occurring in only 4 of the 21 combinations: GPT-4.1 mini on Summarization (9.79→9.05) and Reasoning (26.40→18.45), and TigerLLM-9B on Summarization (26.17→17.99) and Reasoning (53.45→30.20). Several apparent gains do not constitute genuine improvement. Mistral-nemo-12B on GQA (55.42→49.60)

Model	Task	Before CoT			After CoT		
		A-err.	B-err.	BHS	A-err.	B-err.	BHS
DeepSeek-R1-14B	GQA	48.00	29.20	38.60	75.05	16.43	45.74 ↑
	Summarization	17.60	6.73	12.17	20.70	9.07	14.89 ↑
	Reasoning	19.90	16.30	18.10	24.40	19.70	22.05 ↑
GPT-4.1 mini	GQA	13.90	17.23	15.56	38.90	34.65	36.78 ↑
	Summarization	8.20	11.37	9.79	6.30	11.80	9.05 ↓
	Reasoning	31.70	21.10	26.40	11.60	25.30	18.45 ↓
Qwen2.5-32B	GQA	27.90	65.80	46.85	74.40	23.25	48.83 ↑
	Summarization	4.70	10.73	7.72	33.60	7.47	20.54 ↑
	Reasoning	24.80	28.80	26.80	8.70	49.10	28.90 ↑
Gemma-2-27B	GQA	16.40	78.50	47.45	50.50	38.58	44.54 ↓
	Summarization	4.20	31.40	17.80	19.00	11.03	15.02 ↓
	Reasoning	5.50	72.60	39.05	3.60	82.30	42.95 ↑
Mistral-nemo-12B	GQA	85.90	24.93	55.42	93.30	5.90	49.60 ↓
	Summarization	70.50	9.87	40.19	45.40	14.10	29.75 ↓
	Reasoning	1.60	97.00	49.30	0.10	99.80	49.95 ↑
LLaMA-3.1-8B	GQA	13.80	78.05	45.93	83.30	12.90	48.10 ↑
	Summarization	31.80	25.77	28.79	64.90	21.80	43.35 ↑
	Reasoning	17.40	89.20	53.30	28.70	75.60	52.15 ↓
TigerLLM-9B	GQA	18.40	38.17	28.28	21.80	38.37	30.09 ↑
	Summarization	4.80	47.53	26.17	17.40	18.57	17.99 ↓
	Reasoning	21.40	85.50	53.45	8.40	52.00	30.20 ↓

Table 6: BenHalluScore (BHS) before and after Chain-of-Thought (CoT) prompting. A-err. = false alarm rate (%); B-err. = miss rate (%); BHS = $\frac{1}{2}(A\text{-err.} + B\text{-err.})$; lower is better. improved (↓); degraded (↑).

and Summarization (40.19→29.75) show falling Track B error but Track A error simultaneously rises above 90%, indicating the model has simply swapped its uniform “No” posture for a uniform “Yes” posture. Among reasoning-oriented models, GPT-4.1 mini improves markedly on the Reasoning task (26.40→18.45), while DeepSeek-R1-14B degrades across all three tasks, suggesting that models with built-in chain-of-thought mechanisms may experience prompt interference rather than benefit from additional CoT instruction. TigerLLM-9B records the largest single improvement across all 21 combinations on the Reasoning task (53.45→30.20), indicating that Bengali-centric pretraining enables CoT to activate more effectively on

structured mathematical tasks. Conversely, Qwen2.5-32B, the best zero-shot performer on Summarization, degrades sharply under CoT (7.72→20.54), with Track A error rising from 4.70% to 33.60%, suggesting that CoT disrupts its already well-calibrated contextual reasoning. The Reasoning task shows the clearest pattern of improvement, particularly for GPT-4.1 mini and TigerLLM-9B, while GQA shows predominantly negative effects across 6 of 7 models, consistent with (Cheng et al., 2025), who find that CoT gains are confined to tasks requiring structured, verifiable intermediate steps.

6 Discussion

Necessity of dual-track calibration.

Single-track frameworks, including AraHalluEval (Alansari and Luqman, 2025), assess models only on hallucinated candidates, leaving unexamined whether a model recognises correct outputs as non-hallucinated. BenHalluEval addresses this by introducing Track A alongside Track B, and aggregating both into BenHalluScore. Our results justify this design — models such as Mistral-nemo-12B appear competitive on Track B alone, yet collapse entirely on Track A, a failure mode that single-track evaluation would conceal. The CoT mitigation results further reinforce this necessity: in several cases, CoT reduces Track B error while simultaneously driving Track A error above 90%, a bias shift that BenHalluScore captures directly but a single-track metric would reward as improvement.

GQA vs. Code-Mixed QA. The Bangla-English Code-Mixed QA task shares its hallucination taxonomy and candidate instances with GQA, differing only in question script (Bengali to Bangla-English). Despite this structural equivalence, BenHalluScores diverge across both tasks (Table 4). Most multilingual models achieve better results on Bangla-English Code-Mixed QA, whereas TigerLLM-9B and GPT-4.1 mini perform more favourably on Bengali GQA. This indicates that the input script has a measurable influence on hallucination judgement even when the underlying content is identical.

Language-specific advantage. Despite having the fewest parameters in our suite, TigerLLM-9B achieves competitive BenHalluScores on GQA and Summarization, outperforming several larger multilingual models. We attribute this to Bangla-centric pretraining rather than scale, suggesting that language-specific optimisation is a more reliable predictor of Bangla hallucination detection performance than model size. This advantage extends to CoT mitigation: TigerLLM-9B records the largest single BenHalluScore improvement across all 21 model-task combinations on Reasoning (53.45→30.20), further suggesting that Bengali-oriented pretraining enables more

effective activation of structured reasoning under CoT instruction.

Limits of prompting-based mitigation.

CoT prompting gives divergent outcomes across the 21 model-task combinations examined. Nine combinations show a reduction in BenHalluScore after CoT while twelve worsen. The effect is task-sensitive: Reasoning shows the highest rate of genuine improvement, particularly for GPT-4.1 mini and TigerLLM-9B, while GQA shows predominantly negative effects, with 6 of 7 models degrading. This pattern is consistent with (Cheng et al., 2025), who find that CoT gains are confined to tasks requiring structured, verifiable intermediate steps - a condition more closely met by mathematical reasoning than by open-ended QA or source-grounded summarization. Furthermore, reasoning-oriented models do not uniformly benefit from explicit CoT instruction: DeepSeek-R1-14B degrades across all three tasks, suggesting that models with built-in chain-of-thought mechanisms may experience prompt interference when CoT is applied externally. Response biases persist across model categories, and prompting strategies alone are insufficient as a hallucination mitigation approach in Bengali.

7 Conclusion

Reliable hallucination evaluation for Bengali has remained absent despite the language’s global reach. We introduced BenHalluEval, a multi-task hallucination evaluation framework covering four tasks with 12,000 hallucinated candidates, evaluated across seven LLMs under a dual-track protocol. We proposed BenHalluScore, a calibration metric that jointly penalises false positives and missed detections, preventing inflated scores from uniform response bias. BenHalluScore ranges from 7.72% to 55.42% across models and tasks, with Bengali-centric pretraining proving a stronger predictor of calibration quality than parameter count and input script measurably influencing model behaviour. Chain-of-thought prompting, tested across all seven models on three tasks, improves performance in 9 of 21 combinations but degrades it in 12, with benefits confined to the Reasoning task and predominantly negative effects on GQA, confirming

that prompting-only mitigation is insufficient for consistent hallucination discrimination in Bengali. Future work should explore few-shot protocols, consistency-based mitigation strategies, and extensions of the dual-track design to other low-resource languages.

Limitations

While we conduct manual validation for selected components of the benchmark, the hallucinated candidates are generated automatically using GPT-5.4 rather than fully authored by human annotators. As a result, the benchmark may still reflect generation-specific patterns that differ from naturally occurring hallucinations in deployed systems. BenHalluScore assigns equal weight to errors on ground-truth and hallucinated tracks to reduce the influence of systematic response tendencies, but deployment settings with asymmetric costs may require task-specific weighting. All models are evaluated zero-shot at temperature 0, so results may differ under few-shot prompting or model-specific prompt tuning. The chain-of-thought mitigation experiment covers three tasks but does not include Code-Mixed QA, leaving the interaction between code-mixed input script and step-by-step prompting unexplored. Finally, BenHalluEval is scoped to four Bengali and Bangla-English task settings; generalization to other low-resource languages, Bengali genres, domains, and more complex mathematical reasoning remains to be validated.

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References

- Samir Abdaljalil, Parichit Sharma, Erchin Serpedin, and Hasan Kurban. 2026. [Halluverse-M³: A multitask multilingual benchmark for hallucination in LLMs](#). *arXiv preprint arXiv:2602.06920*.
- Shefayat E Shams Adib, Ahmed Alfey Sani, Ekramul Alam Esham, Ajwad Abrar, and Tareque Mohmud Chowdhury. 2025. [Assessing large language models for medical qa: Zero-shot and llm-as-a-judge evaluation](#). In *2025 28th International Conference on Computer and Information Technology (ICCIT)*, pages 3081–3086. IEEE.
- Kazi Samin Yasar Alam, Md Tanbir Chowdhury, Tamim Ahmed, Ajwad Abrar, and Md Rafid Haque. 2026. Mixsarc: A bangla-english code-mixed corpus for implicit meaning identification. *arXiv preprint arXiv:2602.21608*.
- Sadia Alam, Md Farhan Ishmam, Navid Hasin Alvee, Md Shahnewaz Siddique, Md Azam Hosain, and Abu Raihan Mostofa Kamal. 2025. Bnsentmix: A diverse bengali-english code-mixed dataset for sentiment analysis. In *Proceedings of the First Workshop on Language Models for Low-Resource Languages*, pages 68–77.
- Aisha Alansari and Hamzah Luqman. 2025. [Ara-HalluEval: A fine-grained hallucination evaluation framework for Arabic LLMs](#). *arXiv preprint arXiv:2509.04656*.
- Yejin Bang, Ziwei Ji, Alan Schelten, Anthony Hartshorn, Tara Fowler, Cheng Zhang, Nicola Cancedda, and Pascale Fung. 2025. [HalluLens: LLM hallucination benchmark](#). *arXiv preprint arXiv:2504.17550*.
- Abhik Bhattacharjee, Tahmid Hasan, Wasi Ahmad, Kazi Samin Mubasshir, Md Saiful Islam, Anindya Iqbal, M. Sohel Rahman, and Rifat Shahriyar. 2022. [BanglaBERT: Language model pretraining and benchmarks for low-resource language understanding evaluation in Bangla](#). In *Findings of the Association for Computational Linguistics: NAACL 2022*, pages 1318–1327, Seattle, United States. Association for Computational Linguistics.
- Cl  a Chataigner, Afaf Ta  ik, and Golnoosh Farnadi. 2024. Multilingual hallucination gaps in large language models. *arXiv preprint arXiv:2410.18270*.
- Jiahao Cheng, Tiancheng Su, Jia Yuan, Guoxiu He, Jiawei Liu, Xinqi Tao, Jingwen Xie, and Huaxia Li. 2025. Chain-of-thought prompting obscures hallucination cues in large language models: An empirical evaluation. *arXiv preprint arXiv:2506.17088*.
- Jonathan H. Clark, Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. 2020. [TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages](#). *Transactions of the Association for Computational Linguistics*, 8:454–470.
- Tahmid Hasan, Abhik Bhattacharjee, Md. Saiful Islam, Kazi Mubasshir, Yuan-Fang Li, Yong-Bin Kang, M. Sohel Rahman, and Rifat Shahriyar.

2021. [XL-sum: Large-scale multilingual abstractive summarization for 44 languages](#). In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 4693–4703, Online. Association for Computational Linguistics.
- Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2024. [A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions](#). *ACM Transactions on Information Systems*.
- Saad Obaid ul Islam, Anne Lauscher, and Goran Glavaš. 2025. [How much do LLMs hallucinate across languages? on realistic multilingual estimation of LLM hallucination](#). *arXiv preprint arXiv:2502.12769*.
- Saman Sarker Joy and Swakkhar Shatabda. 2025. Bnmmlu: Measuring massive multitask language understanding in bengali. *arXiv preprint arXiv:2505.18951*.
- Alvi Khan, Fida Kamal, Mohammad Abrar Chowdhury, Tasnim Ahmed, Md Tahmid Rahman Laskar, and Sabbir Ahmed. 2023. [BanglaCHQ-summ: An abstractive summarization dataset for medical queries in Bangla conversational speech](#). In *Proceedings of the First Workshop on Bangla Language Processing (BLP-2023)*, pages 85–93, Singapore. Association for Computational Linguistics.
- Junyi Li, Xiaoxue Cheng, Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. 2023. [HaluEval: A large-scale hallucination evaluation benchmark for large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6449–6464, Singapore. Association for Computational Linguistics.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. [TruthfulQA: Measuring how models mimic human falsehoods](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland. Association for Computational Linguistics.
- Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. [FActScore: Fine-grained atomic evaluation of factual precision in long form text generation](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore. Association for Computational Linguistics.
- Shrey Pandit, Jiawei Xu, Junyuan Hong, Zhangyang Wang, Tianlong Chen, Kaidi Xu, and Ying Ding. 2025. Medhallu: A comprehensive benchmark for detecting medical hallucinations in large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 2858–2873.
- Bidyarthi Paul, Jalisha Jashim Era, Mirazur Rahman Zim, Tahmid Sattar Aothoi, and Faisal Muhammad Shah. 2025. Leveraging large language models for bengali math word problem solving with chain of thought reasoning. *arXiv preprint arXiv:2505.21354*.
- Forrest Ping and Greg Durrett. 2024. [FaithBench: A diverse hallucination benchmark for summarization by modern LLMs](#). *arXiv preprint arXiv:2410.13210*.
- Ahmed Rafid, Rumman Adib, Fariya Ahmed, Ajwad Abrar, and Mohammed Saidul Islam. 2026. Banglasummeval: Reference-free factual consistency evaluation for bangla summarization. In *Proceedings of the Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026)*, pages 595–608.
- Nishat Raihan and Marcos Zampieri. 2025. Tigerllm-a family of bangla large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 887–896.
- Ahmed Alfey Sani, Kazi Akib Zaoad, Shefayat E Shams Adib, Md Abdul Muqtadir, and Ajwad Abrar. 2026. [Addressing data scarcity in bangla fake news detection: An llm-based dataset augmentation approach](#). *Preprint, arXiv:2605.01292*.
- Rajvee Sheth and 1 others. 2025. [Beyond monolingual assumptions: A survey of code-switched NLP in the era of large language models across modalities](#). *arXiv preprint arXiv:2510.07037*.
- Mahbub E Sobhani, Md. Faiyaz Abdullah Sayeedi, Tasnim Mohiuddin, Md Mofijul Islam, and Swakkhar Shatabda. 2025. [MathMist: A parallel multilingual benchmark dataset for mathematical problem solving and reasoning](#). *arXiv preprint arXiv:2510.14305*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35.
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. [BERTscore: Evaluating text generation with BERT](#). In *International Conference on Learning Representations*.

A Hallucination Type Definitions and Examples

Table 7 lists all hallucination types used across the three generation tasks, with a definition and a translated English example for each. Bangla-English Code-Mixed QA shares the same four types as GQA and is omitted here for brevity.

B Benchmark Statistics by Hallucination Type

Table 8 reports the number of hallucinated candidates generated per type and task, totalling 12,000 instances across the full benchmark.

C Hallucination Generation Prompts

The following prompts were used to generate hallucinated candidates via GPT-5.4. Each prompt specifies a hallucination method, includes a demonstration, and formats the target instance with field markers. Demonstrations are shown in Bengali for GQA and Reasoning, and in Latin-script Bangla-English for the Code-Mixed task.

C.1 GQA Generation Prompt

Table 9 shows the full GQA generation prompt with Bengali demonstrations for all four methods.

C.2 Code-Mixed QA Generation Prompt.

Table 10 mirrors the GQA prompt but instructs the model to generate hallucinated answers in Latin-script Bangla-English throughout.

C.3 Summarization Generation Prompt.

Table 11 shows the summarization generation prompt covering three hallucination methods.

C.4 Reasoning Generation Prompt

Table 12 shows the reasoning generation prompt. GPT-5.4 is instructed to modify exactly one step and return output as a structured JSON object.

C.5 Bengali-to-Bangla-English Conversion Prompt

Table 13 shows the Bengali-to-Bangla-English conversion prompt used to produce the code-mixed seed instances from the GQA questions.

D Hallucination Evaluation Prompts

The following prompts were used to elicit binary hallucination verdicts (*yes/no*) from the evaluated models. Zero-shot prompts request a single-token response; chain-of-thought prompts require step-by-step reasoning before the final verdict. All prompts are in English except for the input fields, which are in Bengali or Bangla-English depending on the task.

D.1 Zero-Shot Evaluation Prompts

Tables 14, 15, and 16 show the zero-shot evaluation prompts for GQA/Code-Mixed QA, Summarization, and Reasoning respectively.

D.2 Chain-of-Thought Evaluation Prompts

Under the CoT condition, models are asked to reason through three steps before issuing a verdict. Tables 17, 18, and 19 show the corresponding prompts for GQA/Code-Mixed QA, Summarization, and Reasoning respectively.

E Full Per-Task Evaluation Results

The following tables extend Table 4 in the main paper by reporting raw verdict counts alongside error rates for all seven models across each of the four tasks. A-err.: Track A error rate on ground-truth instances. B-err.: Track B error rate on hallucinated candidates. BHS: BenHalluScore = $\frac{1}{2}(A\text{-err.} + B\text{-err.})$.

E.1 Generative Question Answering

Table 20 shows the full GQA evaluation results across all seven models under the dual-track BanglaHalluScore.

E.2 Summarization

Table 21 shows the full summarization evaluation results across all seven models under the dual-track BanglaHalluScore.

E.3 Reasoning

Table 22 shows the full mathematical reasoning evaluation results across all seven models under the dual-track BanglaHalluScore.

E.4 Bangla-English Code-Mixed QA

Table 23 shows the full Code-Mixed Banglish QA evaluation results across all seven models under the dual-track BanglaHalluScore.

E.5 GQA Model Error Distribution

Figure 3 visualises Track A error, Track B error, and BenHalluScore (BHS) for all seven models on GQA, sorted by ascending BHS. The chart illustrates the contrasting bias profiles across model categories: GPT-4.1 mini achieves the best balance (BHS 15.56%), while Mistral-nemo-12B scores worst (BHS 53.84%), driven by its 87.70% Track A error. Models such as LLaMA-3.1-8B show the opposite pattern, with high Track B error and low Track A error, reflecting a systematic “Yes” bias rather than genuine hallucination discrimination.

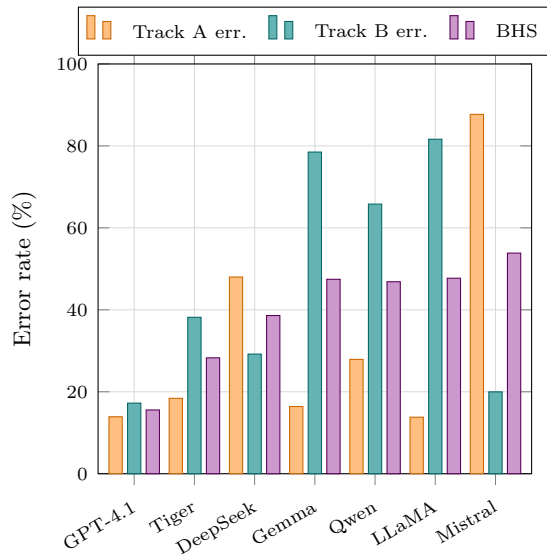


Figure 3: Track A error, Track B error, and BenHalluScore (BHS) for all seven models on GQA, sorted by ascending BHS. GPT-4.1 mini achieves the best balance with the lowest BHS (15.56%). Mistral-nemo-12B scores worst (53.84%), driven by its 87.70% Track A error.

F Hallucination Indicators

Table 24 lists the full set of hallucination indicators used during annotation and quality

review. Indicators marked with * are Bangla-specific and do not appear in prior English or Arabic hallucination benchmarks.

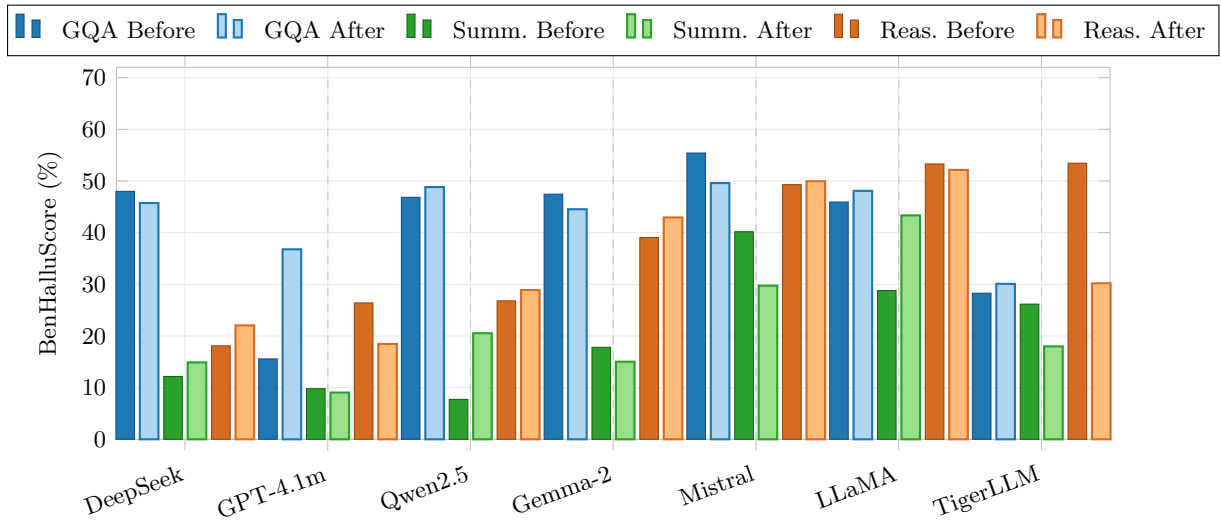


Figure 4: BenHalluScore before and after CoT prompting across all seven models and three tasks (21 model-task combinations). Darker bars = Before CoT; lighter bars = After CoT. Lower is better. Blue = GQA; Green = Summarization; Orange = Reasoning. An *After* bar shorter than its *Before* bar indicates improvement.

Task	Type	Definition	Example
GQA	Factualness	Fabricates a concrete fact (name, date, number, place) not present in the context.	Q: When was the Padma Bridge inaugurated? Ground-Truth: 2022 Hallucinated: The Padma Bridge was inaugurated in 2019.
	Comprehension	Misunderstands the question context or intention.	Q: What is the capital of Bangladesh? Ground-Truth: Dhaka Hallucinated: Bangladesh is a South Asian country.
	Specificity	Answers at an inappropriate level of detail.	Q: How many players are in a cricket team? Ground-Truth: 11 Hallucinated: Some.
	Inference	Draws a conclusion not supported by the given information.	Q: Where is Cox’s Bazar located? Ground-Truth: Chittagong Division Hallucinated: Cox’s Bazar is the capital of Bangladesh.
Summ.	Fabricated Content	Introduces a detail not present anywhere in the source.	Context: Patient reports mild fever for two days. Hallucinated: Patient has been hospitalised for a week.
	Non-Factual Addition	Inserts a plausible but unverifiable claim into the summary.	Context: Child has a recurring cough. Hallucinated: Child likely has bronchial asthma.
	Direct Contradiction	Reverses information explicitly stated in the source.	Context: Pain improves when walking. Hallucinated: Pain does not improve with walking.
Reasoning	Arithmetic Slip	A single numerical computation is silently altered.	Correct: $25 \times 8 = 200$ hours Hallucinated: $25 \times 8 = 180$ hours
	Formula Misapplication	A related but incorrect operation is substituted.	Correct: $200 \times 5 = 1000$ Hallucinated: $200 + 5 = 205$
	Invalid Deduction	A logically unsupported transition is injected between steps.	Correct: Tax = $20000 \times 0.1 = 2000$ Hallucinated: Tax is split per pair of employees.
	Hallucinated Intermediate Fact	A fabricated assumption is introduced mid-chain as if given.	Correct: Working days = 25 Hallucinated: Workers take 2 days off, so working days = 23.
	Semantic Drift	The reasoning drifts to answer a different question than asked.	Correct: Total cost = wages + tax Hallucinated: Total = wages only = \$20,000.

Table 7: Hallucination type definitions and examples across GQA, Summarization, and Reasoning. Bengali examples are translated to English for readability.

Task	Hallucination type	Count
GQA	Factualness	1,000
	Comprehension	1,000
	Specificity	1,000
	Inference	1,000
Code-mix	Factualness	1,000
	Comprehension	1,000
	Specificity	1,000
	Inference	1,000
Summ.	Fabricated content	1,000
	Non-factual addition	1,000
	Direct contradiction	1,000
Reasoning	Arithmetic slip	200
	Formula misapplication	200
	Invalid deduction	200
	Hallucinated intermediate fact	200
	Semantic drift	200
Total		12,000

Table 8: Hallucinated candidate counts by task and type.

[System instruction] I want you act as a hallucination answer generator. The answer should be given in BANGLA. Given a question, right answer, and related knowledge, your objective is to write a hallucinated answer that sounds plausible but is factually incorrect using one of the following methods: Method 1 — Factualness: You are trying to answer a question but there is a factual contradiction between the answer and the knowledge. You can fabricate some information that does not exist in the provided knowledge. Demonstration: #Knowledge#: “উইলিয়াম আব্রাহাম সাইমন ওডারল্যান্ড (জন্ম: ৬ ডিসেম্বর, ১৯১৭ — মৃত্যু: ১৮ই মে, ২০০১) ছিলেন একজন ওলন্দাজ-অস্ট্রেলীয় সামরিক কমান্ডো অফিসার...” #Q#: “উইলিয়াম কবে জন্মগ্রহণ করেন?” #Gold#: “৬ ডিসেম্বর, ১৯১৭” #Hallucinated#: “৬ নভেম্বর, ১৯১৬” Method 2 — Comprehension: You are trying to answer a question but you misunderstand the question context and intention. Demonstration: #Q#: “স্বাধীন বাংলাদেশের প্রথম চলচ্চিত্রটির নাম কী?” #Gold#: “সুকুমারী” #Hallucinated#: “জহির রায়হান” Method 3 — Specificity: You are trying to answer a question but the answer is too general or too specific. Demonstration: #Q#: “কুয়েটের মোট ছাত্রছাত্রীর সংখ্যা কত?” #Gold#: “প্রায় ৬ হাজার” #Hallucinated#: “অজানা” Method 4 — Inference: You are trying to answer a question but the answer cannot be inferred from the knowledge. Demonstration: #Q#: “ঢাকার মোট আয়তন কত?” #Gold#: “১৩৪ বর্গমাইল” #Hallucinated#: “২০ মিলিয়ন জনসংখ্যা” #Hallucinated Answer# can only have about 5 more words than #Right Answer#. #Knowledge#: (context) #Question#: (question) #Right Answer#: (gold) #Hallucinated Answer#: Generate
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Table 9: GQA hallucination generation prompt. One API call per method per seed instance. The same structure is used for Code-Mixed QA with Bangla-English output and demonstrations.

[System instruction] I want you act as a hallucination answer generator. The answer should be given in CODE-MIXED BANGLA (Bangla-English: Latin-script Bengali). Given a question, right answer, and related knowledge, write a hallucinated answer that sounds plausible but is factually incorrect using the four methods below. Method 1 — Factualness: Fabricate information not present in the knowledge. Demo: #Knowledge#: “William Abraham Simon Ouderland (janma: 6 december, 1917, mrittu: 18 may, 2001) chhilen ekjon commando officer.” #Q#: “Ouderland kobe janmogrohon koren?” #Gold#: “6 december, 1917” #Hallucinated#: “6 november, 1916” Method 2 — Comprehension: Misunderstand the question context and intention. Demo: #Q#: “Swadhin Bangladesher prothom cholochchitrotir nam ki?” #Gold#: “Sukumari” #Hallucinated#: “Jahir Raihan” Method 3 — Specificity: Answer at an inappropriate level of specificity. Demo: #Q#: “bortomane kuet-er mot chatrochhatrir shongkha koto?” #Gold#: “pray 6 hajar” #Hallucinated#: “ojana” Method 4 — Inference: Draw a conclusion not supported by the knowledge. Demo: #Q#: “Dhakar mot ayoton koto?” #Gold#: “134 borgomail” #Hallucinated#: “20 million jonoshon-kha” #Hallucinated Answer# can only have about 5 more words than #Right Answer#. #Knowledge#: (context) #Question#: (question) #Right Answer#: (gold) #Hallucinated Answer#: Generate
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Table 10: Code-Mixed QA hallucination generation prompt. Four methods mirror the GQA prompt but use Latin-script Bangla-English throughout.

[System instruction] I want you act as a hallucination summary generator. The summary should be given in BANGLA. Given a document and the right summary, write a hallucinated summary that sounds plausible but is factually incorrect using one of the following methods: Method 1 — Fabricated Content: You fabricate information that does not exist in the document. Demo: #Doc#: “আমার মেয়ের বয়স ৫ বছর। সর্দি সমস্যা।” #Gold#: “বয়স ৫, সর্দি সমস্যা, করণীয় কী” #Hallucinated#: “বয়স ৫, জ্বর ও কাশি সহ হাসপাতালে ভর্তি প্রয়োজন” Method 2 — Non-Factual Addition: Insert information that cannot be inferred from the document. Demo: #Doc#: “স্ট্রীর বয়স ২২, ডেট ৭ নভেম্বর থেকে ১৪ তারিখে পরিবর্তন।” #Gold#: “স্ট্রীর বয়স ২২, ডেট পরিবর্তন” #Hallucinated#: “ডাক্তার জুলাই মাসে ডেট দিয়ে পরে ১৪ করেছে” Method 3 — Direct Contradiction: Introduce a clear contradiction between summary and document. Demo: #Doc#: “হাঁটলে ব্যথা ঠিক হয়ে যায়।” #Gold#: “হাঁটলে ঠিক হয়” #Hallucinated#: “হাঁটলেও ব্যথা কমে না” #Hallucinated Summary# can only have about 5 more words than #Right Summary#. #Document#: (doc) #Right Summary#: (gold) #Hallucinated Summary#: Generate
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Table 11: Summarization hallucination generation prompt. Three methods correspond to fabricated content, non-factual addition, and direct contradiction.

[System instruction]

I want you to act as a hallucinated reasoning chain generator. The output must be in **BANGLA**. Given a math problem, a correct step-by-step reasoning chain, and the correct final answer, your objective is to produce a hallucinated reasoning chain that sounds plausible but is mathematically incorrect. You MUST use one of the following hallucination methods:

Method 1 — Arithmetic Slip:

Keep the logic structure completely intact but insert one subtle numerical error in a single calculation step. The wrong number should be close to the correct one (roughly 5% to 20% off). All subsequent steps must use the wrong number consistently. Preserve the `<<expr=result>>` annotation format.

Demonstration:

#Correct step#: “২৫ দিন * ৮ ঘন্টা/দিন = `<<২৫*৮=২০০>>২০০ ঘন্টা`”

#Hallucinated step#: “২৫ দিন * ৮ ঘন্টা/দিন = `<<২৫*৮=১৮০>>১৮০ ঘন্টা`”

Method 2 — Formula Misapplication:

Use a related but wrong operation in one step. The Bengali narrative must justify the wrong operation naturally. All subsequent steps must be consistent with the wrong result.

Demonstration:

#Correct step#: “২০০ ঘন্টা * ৫/ঘন্টা = `<<২০০*৫=১০০০>>১০০০`”

#Hallucinated step#: “২০০ + ৫ = `<<২০০+৫=২০৫>>২০৫`”

Method 3 — Invalid Deduction:

Inject one logically unsupported conclusion in a single step. All subsequent steps must follow from this invalid conclusion.

Demonstration:

#Correct step#: “৳২০,০০০ * .১ = `<<২০০০০*.১=২০০০>>২,০০০`”

#Hallucinated step#: “যেহেতু কর্মচারীর সংখ্যা জোড় (৬ জন), তাই প্রতিটি জুটির জন্য আলাদাভাবে হিসাব করতে হবে। প্রতি জুটির মজুরি = $৳২০,০০০ \div ৩ = <<২০০০০/৩=৬৬৬৭>>৳৬,৬৬৭।$ ”

Method 4 — Hallucinated Intermediate Fact:

Introduce one fabricated assumption not present in the problem as if it naturally follows. All subsequent steps must proceed from this fabricated fact.

Demonstration:

#Correct step#: “২৫ দিন * ৮ ঘন্টা/দিন = `<<২৫*৮=২০০>>২০০ ঘন্টা`”

#Hallucinated step#: “কমীরা প্রতি মাসে ২ দিন ছুটি ভোগ করেন, তাই কার্যকর কাজের দিন = $২৫ - ২ = <<২৫-২=২৩>>২৩$ দিন। মোট ঘন্টা = $২৩ * ৮ = <<২৩*৮=১৮৪>>১৮৪$ ঘন্টা।”

Method 5 — Semantic Drift:

Slightly reinterpret what the problem is asking mid-reasoning so the final answer addresses a different question. The drift must feel like a plausible reading of the problem.

Demonstration:

#Correct step#: “৳২,০০০ + ৳২০,০০০ = `<<২০০০+২০০০০=২২০০০>>২২,০০০`”

#Hallucinated step#: “যেহেতু প্রশ্নটি মূলত মজুরির মোট পরিমাণ জিজ্ঞেস করছে, তাই মোট মজুরি ৳২০,০০০ই চূড়ান্ত উত্তর।”

#Problem#: *<insert Bengali math problem>*

#Correct Reasoning Chain#: *<insert ground-truth CoT>*

#Correct Answer#: *<insert ground-truth answer>*

#Hallucinated Chain#: Generate

Table 12: Hallucination generation prompt for the Reasoning task. Five methods correspond to arithmetic slip, formula misapplication, invalid deduction, hallucinated intermediate fact, and semantic drift.

[System instruction] Convert Bengali QA text into natural Bangla-English — Bengali-English code-mix written entirely in Latin script, the way people write on South Asian social media or in casual chats. Output only Latin script - no Bengali characters anywhere. Output only the converted text. No labels, no explanations.

Rules: (1) Romanise Bengali words naturally; use English words where code-mix speakers naturally would. (2) Do NOT translate everything to English; preserve Bengali phrasing where it feels natural. (3) Preserve meaning, tone, names, numbers, dates, brands, and technical terms exactly.

Spelling variation (apply to ~15% of eligible words): common Bangla-English alternates: *bhai/vai, bhalo/valo, ache/ase, ki/kee, onek/aneek, ektu/aktu, shob/sob, nai/nei, hoise/hoiche*.

Abbreviations (apply sparingly, ~10% of eligible words): *phn* (phone), *msg* (message), *prob* (problem), *pls/plz* (please), *bcz/coz* (because).

Do not: over-abbreviate, use excessive expressive spellings, change factual meaning, or make output hard to read.

[Text to convert]: *<Bengali question and/or passage>*

Table 13: Prompt used to convert the 1,000 GQA seed instances from Bengali to Bangla-English. Applied separately to questions and knowledge passages. Ground-Truth answers are retained in their original Bengali form.

You are an evaluator. Decide whether the provided model answer is hallucinated relative to the context and question. Only reply with a single token: **yes** or **no**. No explanation, no punctuation, no extra text. *yes* means the answer contains information not supported by the context or contradicts it. Provide the answer in English only.

Context: $\langle context \rangle$
Question: $\langle question \rangle$
Model answer: $\langle answer \rangle$

Answer now:

Table 14: Zero-shot evaluation prompt for GQA and Code-Mixed QA. For code-mixed instances, all fields are in Bangla-English.

You are an evaluator. Decide whether the provided summary is hallucinated relative to the document. Only reply with a single word: **yes** or **no**. *yes* means the summary contains information not supported by or contradicting the document. Provide the answer in English only.

Document: *⟨Bengali medical article⟩*

Summary: *⟨candidate summary⟩*

Answer now (yes or no):

Table 15: Zero-shot evaluation prompt for Summarization.

You are an expert evaluator for Bengali mathematical reasoning tasks. Decide whether the provided reasoning chain is hallucinated. Reply with a single token only: **yes** or **no**. *yes* = chain or answer contains incorrect steps or unsupported facts. English only.

Problem: $\langle \text{Bengali math problem} \rangle$
Reasoning Chain: $\langle \text{CoT solution} \rangle$
Final Answer: $\langle \text{final answer} \rangle$

Answer now:

Table 16: Zero-shot evaluation prompt for Reasoning.

You are an evaluator checking whether a model answer is hallucinated.

Question: $\langle question \rangle$ Model Answer: $\langle answer \rangle$

Analyze step by step:

Step 1: What factual claims does the answer make?

Step 2: Are these claims supported by or inferable from the question context?

Step 3: Based on steps 1–2, is the answer hallucinated?

Final answer (write only this word on the last line):
yes or **no**
(yes = hallucinated, no = not hallucinated)

Table 17: Chain-of-thought evaluation prompt for GQA and Code-Mixed QA.

You are an evaluator checking whether a summary is hallucinated relative to a document.

Document: $\langle document \rangle$ **Summary:** $\langle summary \rangle$

Analyze step by step:

Step 1: List the key claims made in the summary.

Step 2: For each claim, check whether it is directly supported by the document.

Step 3: Based on steps 1–2, decide your final answer.

Final answer (write only this word on the last line):
yes or **no**
(yes = hallucinated, no = not hallucinated)

Table 18: Chain-of-thought evaluation prompt for Summarization.

You are an expert evaluator for Bengali mathematical reasoning tasks.

Question: $\langle problem \rangle$ Reasoning Chain: $\langle CoT \rangle$

Answer: $\langle final\ answer \rangle$

Analyze step by step:

Step 1: Is each calculation or logical step mathematically correct?

Step 2: Does the final answer follow logically from the reasoning chain?

Step 3: Based on steps 1–2, is this reasoning chain hallucinated?

Respond **ONLY** with a JSON object on the last line:

```
{"is_hallucinated": "Yes"}
```

 or

```
{"is_hallucinated": "No"}
```

Table 19: Chain-of-thought evaluation prompt for Reasoning.

#	Model	Track A — Ground Truth (1,000)			Track B — Hallucinated (4,000)			BHS
		Correct	Wrong	A-err.	Correct	Wrong	B-err.	
1	Qwen2.5-32B-Instruct	771	279	27.90%	1,365	2,632	65.80%	46.85%
2	Gemma-2-27B	836	164	16.40%	860	3,140	78.50%	47.45%
3	DeepSeek-R1-14B	342	480	48.00%	2,820	1,168	29.20%	38.60%
4	GPT-4.1 mini	861	139	13.90%	3,311	689	17.23%	15.56%
5	Mistral-nemo-12B	123	877	87.70%	3,201	799	19.98%	53.84%
6	LLaMA-3.1-8B-Instruct	862	138	13.80%	735	3,265	81.63%	47.71%
7	TigerLLM-9B	816	184	18.40%	2,855	1,145	38.17%	28.28%

Table 20: GQA full results. **Bold** = best and worst BHS.

#	Model	Track A — Ground Truth (1,000)			Track B — Hallucinated (3,000)			BHS
		Correct	Wrong	A-err.	Correct	Wrong	B-err.	
1	Qwen2.5-32B-Instruct	953	47	4.70%	2,678	322	10.73%	7.72%
2	Gemma-2-27B	958	42	4.20%	2,058	942	31.40%	17.80%
3	DeepSeek-R1-14B	824	176	17.60%	2,798	202	6.73%	12.17%
4	GPT-4.1 mini	918	82	8.20%	2,659	341	11.37%	9.79%
5	Mistral-nemo-12B	295	705	70.50%	2,704	296	9.87%	40.19%
6	LLaMA-3.1-8B-Instruct	682	318	31.80%	2,227	773	25.77%	28.79%
7	TigerLLM-9B	952	48	4.80%	1,574	1,426	47.53%	26.17%

Table 21: Summarization full results. **Bold** = best and worst BHS.

#	Model	Track A — Ground Truth (1,000)			Track B — Hallucinated (1,000)			BHS
		Correct	Wrong	A-err.	Correct	Wrong	B-err.	
1	Qwen2.5-32B-Instruct	752	248	24.80%	703	288	28.80%	26.80%
2	Gemma-2-27B	945	55	5.50%	274	726	72.60%	39.05%
3	DeepSeek-R1-14B	801	199	19.90%	837	163	16.30%	18.10%
4	GPT-4.1 mini	683	317	31.70%	789	211	21.10%	26.40%
5	Mistral-nemo-12B	984	16	1.60%	30	970	97.00%	49.30%
6	LLaMA-3.1-8B-Instruct	826	174	17.40%	108	892	89.20%	53.30%
7	TigerLLM-9B	786	214	21.40%	145	855	85.50%	53.45%

Table 22: Reasoning full results. **Bold** = best and worst BHS.

#	Model	Track A — Ground Truth (1,000)			Track B — Hallucinated (4,000)			BHS
		Correct	Wrong	A-err.	Correct	Wrong	B-err.	
1	Qwen2.5-32B-Instruct	826	174	17.40%	2,776	1,224	30.60%	24.00%
2	Gemma-2-27B	802	198	19.80%	2,474	1,526	38.15%	28.98%
3	DeepSeek-R1-14B	596	401	40.10%	3,439	551	13.78%	26.94%
4	GPT-4.1 mini	760	240	24.00%	3,208	792	19.80%	21.90%
5	Mistral-nemo-12B	141	859	85.90%	3,003	997	24.93%	55.42%
6	LLaMA-3.1-8B-Instruct	862	138	13.80%	878	3,122	78.05%	45.93%
7	TigerLLM-9B	739	258	25.80%	2,496	1,491	37.28%	31.54%

Table 23: Bangla-English Code-Mixed QA full results. **Bold** = best and worst BHS.

Indicator	Task(s)	Description
Named entity error	GQA, Summ.	Wrong person, place, or organisation
Value error	GQA, Summ.	Incorrect number, date, or quantity
Factual contradiction	GQA	Contradicts world knowledge
Response conflict	GQA	Internal inconsistency
Grammar error	GQA, Summ.	Meaning-altering mistake
Generic answer	GQA	Avoids committing to a fact
Source conflict	GQA	Merges incompatible facts
Fabrication	Summ.	Content not in source article
Inference error	Summ.	Unjustified conclusion
Context inconsistency	Summ.	Contradicts source directly
Code-switching	GQA, Summ.	Responds in English despite Bengali prompt
Transliteration err.*	All	Incorrect Bengali↔Roman mapping
Computation err.*	Reasoning	Wrong step or answer in CoT

Table 24: Full hallucination indicator set. * = Bengali-specific indicator absent from prior benchmarks.